Paleomagnetic Results From the Lassiter Coast, Antarctica, and a Test for Oroclinal Bending of the Antarctic Peninsula

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Paleomagnetic results from 17 magnetically stable units of Upper Cretaceous ('Andean') plutons and dikes of the Lassiter Coast, on the southern Antarctic Peninsula, define a mean paleomagnetic pole at 87°S, 131°W (α_{95} = 11.5°). This indicates that little latitudinal movement of the southern Antarctic Peninsula has occurred during the past 100 m.y. All magnetically stable intrusives are normally polarized and are believed to have been emplaced during the Late Cretaceous epoch of predominantly normal polarity. There is no evidence of postemplacement remagnetization. The uncertainty in declination at the 95% confidence level is computed for both the Lassiter Coast data and those data available from other Andean sites in the Antarctic Peninsula. Within the limits of uncertainty, data from four localities north of 68°S support the contention of Dalziel et al. (1973) that there has not been any apparent post-Late-Cretaceous oroclinal bending in the northern half of the peninsula. For sites to the south in the Lassiter Coast, the uncertainty in declination, due to steep inclinations, is too large to support reliably or deny any large-scale structural bending.

INTRODUCTION

The Antarctic Peninsula, an S-shaped arc of predominantly Mesozoic sedimentary and volcanic rocks intruded by upper Mesozoic to lower Cenozoic plutons of the 'Andean intrusive series' [*Adie*, 1955; *Craddock*, 1969; *Elliot*, 1975], has long been recognized as a southward continuation of the Andean cordillera. The tectonic history of the Antarctic Peninsula during the breakup of Gondwanaland is not completely understood, although a number of interpretations have been advanced [*Hamilton*, 1966; *Heirtzler*, 1971; *Ford*, 1972; *Elliot*, 1972, 1975; *Dalziel et al.*, 1973; *Dalziel and Elliot*, 1972; *Dalziel*, 1974; *Suarez*, 1976]. Of special interest is the origin of the pronounced curvature of the peninsula, which some investigators attribute to major oroclinal bending.

The timing and nature of oroclinal bending of the peninsula have been studied by examining paleomagnetic directions from Andean rocks at several localities. From such analyses of *Blundell*'s [1962] paleomagnetic data, *Hamilton* [1966] proposed that the northern part of the Antarctic Peninsula has been structurally distorted since Early Cretaceous time. This interpretation is not shared by *Dalziel et al.* [1973], who, after calculating mean paleomagnetic directions from *Blundell*'s [1962] data, reasoned that semiangles of the cones of confidence (α_{95}) about the mean directions exceeded variations in declination. Furthermore, paleomagnetic data from an Andean intrusive body at the northern tip of the peninsula did not indicate rotation of the body about a vertical axis since its emplacement [*Dalziel et al.*, 1973]. Owing to the high latitude of the Antarctic Peninsula since at least the mid-Jurassic, reconstructions such as these are open to question unless, specifically, the uncertainty in declination is considered; such considerations were not made in these previous studies.

Paleomagnetic data from the entire length of the Antarctic Peninsula [*Blundell*, 1962; *Dalziel et al.*, 1973; *Scharnberger*, 1974, and unpublished data, 1974; *Kellogg and Reynolds*, 1974; *Reynolds and Kellogg*, 1974] are now sufficiently abundant to test the theories of oroclinal bending. This paper reports paleomagnetic results from the Lassiter Coast and defines confidence limits on mean declinations from all available data from the Antarctic Peninsula in order to test the hypothetical models of structural bending.

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GEOLOGY OF THE LASSITER COAST

The Lassiter Coast lies at the base of the Antarctic Peninsula, bordering the Weddell Sea between latitudes 73° and 75°S and longitudes 60° and 66°W (Figure 1). Much of the mountainous terrain of the Lassiter Coast region consists of highly folded Upper Jurassic argillaceous sandstone and siltstone (Latady Formation) and overlying or laterally equivalent silicic to intermediate volcanic rocks which have been interpreted as being deposits of a Mesozoic arc system [*Suarez*, 1976]. The Latady and volcanic rocks were strongly folded around north to northeast striking axes sometime between Late Jurassic and Late Cretaceous time. The Jurassic sequence was intruded, partially refolded, and contact-metamorphosed by Upper Cretaceous plutons of diorite to granite composition [*Williams and Rowley*, 1971; *Williams et al.*, 1972; *Rowley*, 1973; *Kellogg and Rowley*, 1974; *Plummer*, 1974; *Rowley and Williams*, 1974]. Eight published K-Ar age determinations obtained from plutonic rocks of the southern and central Lassiter Coast range from 119 m.y. to 95 m.y. [*Mehnert et al.*, 1975; *Rowley et al.*, 1975]. Eight other K-Ar age determinations from plutonic rocks of the northern Lassiter coast and southern Black Coast exhibit a similar range in age [*Rowley et al.*, 1976]. Of four dated plutons studied, three yielded acceptable paleomagnetic results.

Some mafic dikes are bracketed between units dated as 95 and 119 m.y. [*Rowley et al.*, 1975], whereas other undated dikes are assumed to be an end phase of the Upper Cretaceous plutonism. Fifteen mafic dikes were collected for paleomagnetic analysis, and of these, seven yielded acceptable paleomagnetic results. One folded and metamorphosed andesite dike that intrudes the Latady Formation and clearly predates the Late Jurassic to Late Cretaceous period of folding was sampled for paleomagnetic study and found to be magnetically unstable.

PALEOMAGNETIC METHOD

Two hundred and ninety separate hand samples, oriented by magnetic compass, were collected at 64 sites from 47 mapped geologic units: 21 plutons, 25 dikes, and one Upper Jurassic(?) metavolcanic unit. Samples from three of the plutons were collected from mafic inclusions.

Wherever possible, at least three, and in most places at least

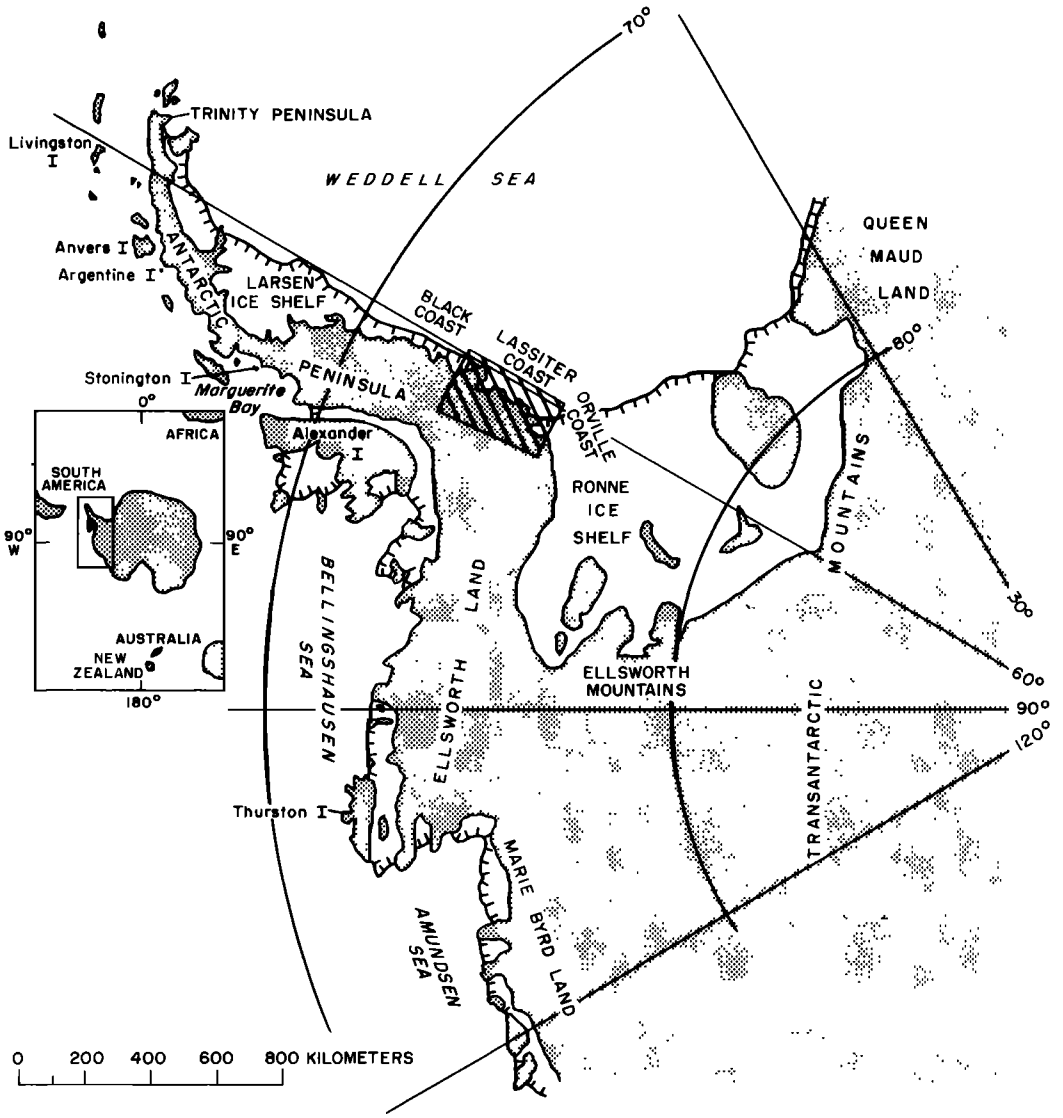


Fig. 1. Index map showing location of the Lassiter Coast, Antarctica. All Lassiter Coast sampling sites occur in the crosshatched area. Extent of shelf ice is shown by hachured line.

five, oriented samples were collected at each outcrop. One oriented core was extracted from each sample in the laboratory. After measurement of the natural remanent magnetization (NRM), the cores were progressively cleaned in peak alternating fields up to 400 Oe. Samples from several plutonic localities were thermally demagnetized to 565°C to compare with the results of af cleaning. Remanent directions and magnetic intensities were measured on a 10-Hz spinner magnetometer.

For the purposes of this study a geological unit is defined as paleomagnetically acceptable if at least three samples were tested and if, at some cleaning level above the NRM, the half angle of the cone of confidence at the 95% confidence level (α_{95}) was less than 25°. The cleaning level yielding the smallest α_{95} was chosen to represent the unit. With only two exceptions, all acceptable sites had at least five samples. A Fisher satistical analysis was applied to the data [*Fisher*, 1953], and a virtual geomagnetic pole (VGP) for each paleomagnetically acceptable locality was calculated (Table 1).

PALEOMAGNETIC RESULTS

Of the 47 geologic units studied, 18 were paleomagnetically acceptable (Table 1): 13 dikes of various composition, 2 plutons, 2 localities of mafic inclusions in plutons, and 1 metamorphosed volcanic rock. Preliminary reports have included many of these data [*Kellogg and Reynolds*, 1974; *Reynolds and Kellogg*, 1974]; however, for a few localities, some refinement of the data has been made by af demagnetization to higher levels. These modifications do not affect the conclusions of the previous papers. Reasons for rejecting the remaining 29 units are listed in Table 2.

All samples from acceptable units are normally magnetized, and in addition, all magnetic directions from rejected but not totally random sites indicate normal magnetization. During demagnetization of all units, no consistent change of direction was observed that might have indicated a major secondary component of magnetization. Three additional tests were made to determine if there were such a component in the rocks:

1. Magnetic intensities were observed following both af demagnetization and thermal demagnetization of selected samples. An increase in remanent intensity of a normally magnetized rock might indicate preferential removal of a secondary (possibly reversed) component as has been observed by *Kellogg* [1971].

2. Thermal demagnetization to 565°C of arbitrarily se-

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TABLE 1. Summary of Paleomagnetic Data, Lassiter Coast, Antarctica

Site and Rock Type	Location	N	Demagnetization Field	$\bar{I}$	$\bar{D}$	K	α_{95}	Paleomagnetic Latitude	Paleomagnetic Longitude	dm	dp
Re21, andesite dike	74.4°S, 63.3°W	4	50	−82	66	17	23	73°S	124°W	45	43
Crowell pluton,* coarse-grained granodiorite	74.4°S, 64.2°W	7	200	−89	64	19	14	75°S	72°W	28	28
Ro192, dacite porphyry dike	74.1°S, 62.8°W	3	50	−81	352	43	19	88°S	53°E	37	36
M206,† granodiorite porphyry dike	74.4°S, 64.8°W	6	100	−85	283	62	9	73°S	27°W	17	17
M307,‡ dacite dike	74.4°S, 64.8°W	5	300	−85	314	23	16	79°S	27°W	32	32
M308,§ coarse-grained granodiorite	74.4°S, 64.8°W	5	150	−69	10	11	24	67°S	111°E	41	34
Ke20, fine-grained mafic dike	73.8°S, 62.8°W	5	400	−86	10	44	12	81°S	71°W	23	23
Ke35, lamprophyre dike	73.5°S, 62.7°W	6	100	−71	83	74	8	54°S	137°W	14	12
Ke39, coarse-grained diorite pluton	73.5°S, 62.7°W	5	200	−84	13	100	8	85°S	95°W	15	15
Ke53, diorite dike	73.1°S, 62.8°W	5	200	−88	91	21	17	73°S	76°W	35	34
Ke62, mafic inclusions in granodiorite	73.2°S, 62.2°W	6	100	−85	42	132	6	79°S	103°W	12	11
Ke71, metabasalt	72.8°S, 63.2°W	5	100	−45	44	85	8	38°S	168°E	11	7
Ro234b, fine-grained mafic dike	73.5°S, 62.9°W	6	200	−76	128	19	16	51°S	98°W	30	27
Ro311c, fine-grained mafic dike	72.8°S, 63.2°W	9	50	−63	346	48	8	60°S	96°E	12	9
Ro314b, fine-grained andesite dike	72.9°S, 62.4°W	6	400	−82	49	714	3	76°S	124°W	5	5
Ro321b, fine-grained mafic dike	72.5°S, 62.9°W	7	50	−66	294	20	14	53°S	30°E	23	19
Bo26, andesite dike	73.4°S, 63.0°W	5	100	−82	6	241	5	88°S	129°W	10	9
Bo67, granodiorite porphyry dike	72.8°S, 63.2°W	6	200	−52	330	72	8	47°S	79°E	11	7
Mean, igneous localities, given equal weight‖	... ...	17	...	...	...	10	11.5	87°S	131°W	...	...

N is number of specimens measured. Demagnetization field, in oersteds, is that at which the smallest α_{95} occurrred. $\bar{I}$ is mean inclination, negative upward. $\bar{D}$ is mean declination, clockwise from true north. K is *Fisher*'s [1953] best estimate of precision; α_{95} is the semiangle of the cone of 95% confidence level; dm and dp are the semimajor and semiminor axes, respectively of the oval of 95% confidence level.

*K-Ar biotite age is 100 ± 2.1 m.y.; K-Ar hornblende age is 98.7 ± 2.1 m.y.

†K-Ar biotite age is 95.2 ± 1.5 m.y.

‡Age bounded by M206 and M308.

§K-Ar biotite age is 104.0 ± 1.6 m.y.

‖Analysis performed directly on VGP positions; VGP of Ke71 not included.

lected samples of rocks from four plutons was undertaken to see if a reversal of direction could be uncovered. Two of these units were stably magnetized; the other two were not. In each case, however, there was no tendency for samples to reverse their magnetic directions.

3. A polished section of one or more samples from each locality was studied under reflected light to detect any discernible regional postemplacement alteration. No petrographic factors could be identified that distinguished acceptable from unacceptable samples, and no characteristics were observed indicating any regional postemplacement alteration. Some samples displayed hematite formation at the expense of magnetite, either as small blebs within or as rims around magnetite grains. About one third of the samples showed various degrees of maghemite formation, indicating low-temperature oxidation. This condition, however, is apparently unrelated to magnetic instability in the rocks that we studied [see *Watkins*, 1967].

The radiometric age dates are consistent with all known geologic relationships, further support for the contention that no significant postemplacement alteration has occurred.

The most plausible explanation for the ubiquitous normal polarity is that plutonism and dike emplacement occurred largely during Late Cretaceous time, when the geomagnetic field was predominantly of normal polarity [*Helsley and Steiner*, 1969; *Keating et al.*, 1975].

Virtual geomagnetic poles from all acceptable units are shown in Figure 2. Fisher analysis was applied to these poles to derive the following paleomagnetic poles. For the 4 stable plutons and the 13 stable dikes an overall paleomagnetic pole at 87.0°S, 130.5°W (α_{95} = 11.5°) was calculated; this is nearly coincident with the present axial dipole field direction. Considered separately, the plutons define a paleomagnetic pole at 87.2°S, 125.7°W (α_{95} = 19.6°), and the dikes define a paleomagnetic pole at 86.5°S, 70.5°E (α_{95} = 13.7°). As a final method of analyzing the data, we considered only the four acceptable localities for which age dates were obtained (refer to Table 1); these sites define a mean VGP at 84.3°S, 5.5°E (α_{95} = 21.6°). The paleomagnetic poles defined by these analyses are all within 5° of each other. This fact, along with the ubiquitous normal polarity, the available age data, and the petrographic and chemical similarities between dikes and plutons [*Kellogg and Rowley*, 1974], leads us to conclude that the mean paleomagnetic pole from all 18 localities adequately defines a Late Cretaceous (approximately 100 m.y.) paleomagnetic pole for the Lassiter Coast.

The VGP for site Ke71, from the metamorphosed Upper Jurassic(?) volcanic sequence, differs markedly from the others (Figure 2). The nearest mapped pluton is 13 km away, although the possibility exists that a pluton closely underlies this locality. Applying a tilt correction about the bedding attitude (N2°W, 55°W) defines a new pole for this site at 11.8°S,

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TABLE 2. List of Rejected Units and Reasons for Rejection

Site and Rock Type*	Location	N	Demagnetization	Reasons
Re1, Re2, Re3, Ro39; granodiorite pluton	74.3°S, 64.0°W	9	NRM to 200	1, 4
Re4; mafic dike	74.3°S, 64.0°W	3	200	1
Re13; granodiorite pluton	74.3°S, 63.9°W	3	200	1
Re15; quartz diorite plug	74.4°S, 63.3°W	6	NRM	4
Re16; porphyry dike	74.4°S, 63.3°W	5	200	1
Ro21; mafic dike	74.3°S, 63.6°W	2	200	3
Re27, Re28; quartz diorite pluton	74.4°S, 63.4°W	17	200	2
Ro95; mafic dike	74.4°S, 64.0°W	9	200	1
Re71, Ro129; granodiorite pluton and baked sediments	74.3°S, 63.4°W	7	NRM to 200	1, 4
Ro150, Re73; granodiorite pluton	74.3°S, 63.1°W	4	200	1
Ro75; quartz diorite pluton	74.3°S, 64.2°W	2	200	1, 3
Re51, Re56; granodiorite pluton	74.0°S, 62.6°W	8	200	1, 4
Re76; quartz diorite pluton	74.3°S, 63.0°W	2	200	3
Re89; quartz diorite pluton	74.2°S, 62.9°W	3	150	1
Re95; quartz diorite pluton	74.2°S, 62.9°W	5	150	1
M208; pluton†	74.4°S, 65.0°W	5	200	1
M414; mafic dike	74.4°S, 65.0°W	3	200	1
Ke3; mafic dike	73.8°S, 64.0°W	6	200	1
Ke8; andesite dike	74.0°S, 63.0°W	4	200	1
Ke31; fine-grained diorite	73.4°S, 63.3°W	6	200	1
Ke42; serpentinized gabbro dike	73.4°S, 62.4°W	5	NRM	4
Ke45; lamprophyre dike	73.5°S, 62.4°W	1	200	3
Ke52; mafic inclusions in quartz diorite pluton	73.1°S, 62.8°W	5	NRM	4
Ke69; diorite pluton	73.3°S, 62.4°W	5	400	1
Ro287c; diorite border facies	73.1°S, 63.3°W	6	NRM	4
Ro317a; mafic dike	72.7°S, 62.5°W	8	200	1
Bo31; mafic inclusions in quartz diorite pluton	73.4°S, 62.7°W	2	200	1, 3
Bo38; gabbro dike	73.5°S, 62.4°W	5	400	2
Ka232; mafic dike	73.5°S, 62.5°W	5	100	1

N is number of samples measured. Demagnetization is the highest level, in oersteds, to which a sample was demagnetized. Reasons for rejection are as follows: 1, unstable magnetization; 2, random stable magnetization; 3, too few samples; 4, intensity of magnetization below resolving capability of magnetometer during NRM measurement.

*Field designation only.

†K-Ar age of biotite is 95.6 ± 5 m.y.

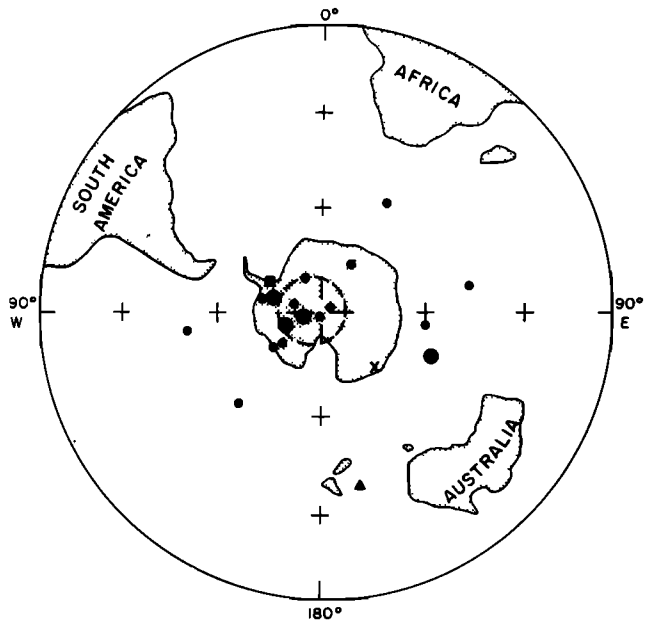


Fig. 2. Virtual geomagnetic poles, Lassiter Coast. The large dots represent VGP's from plutons; the small dots VGP's from dikes; triangle VGP from metabasalt locality Ke71; square Lassiter Coast collecting area; cross present position of the south magnetic pole; dashed circle the 95% circle of confidence (α_{95} = 11.5°) surrounding the mean pole for all intrusive sites (87°S, 131°W).

22.3°W. This position is still markedly divergent from the other VGP's. The thermal and magnetic history for this rock is complex inasmuch as folding and metamorphism occurred prior to or during Late Cretaceous time. This VGP was not included in calculating the paleomagnetic pole for the Lassiter Coast.

OTHER PALEOMAGNETIC RESULTS FROM WEST ANTARCTICA

Other paleomagnetic results from the Antarctic Peninsula consist of data from 12 intrusives in the vicinity of the Argentine Islands, Stonington Island, and Trinity Peninsula [*Blundell*, 1962] and from two localities in a 40-m.y.-old pluton on Livingston Island [*Dalziel et al.*, 1973]. A paleomagnetic pole at 87.8°S, 12.3°E (α_{95} = 6.0°), very near the present geographic South Pole, is derived from Blundell's 12 sites. A similar VGP (82°S, 30°E) is defined by the results of *Dalziel et al.* [1973].

Additional paleomagnetic data from the northern Antarctic Peninsula are given by *Scharnberger* [1974]. He noted that 18 Upper Cretaceous to lower Tertiary intrusives from Anvers Island and the Marguerite Bay area define a mean paleomagnetic pole at 76°S, 33°W (α_{95} = 7°). Four dikes from Stonington Island yield a VGP at 51°S, 26°E (α_{95} = 12°). Most rocks on Stonington Island, however, are considered to be Paleozoic

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or early Mesozoic [*Fraser,* 1965; *Grimley,* 1966; *Craddock,* 1969; *Scharnberger and Scharon,* 1978].

Scharnberger (cited by *McElhinny* [1973, p. 307]) also reports paleomagnetic results from plutons and dikes in the Jones Mountains and Thurston Island of western Ellsworth Land. This region is composed of Paleozoic intrusives and high-grade metamorphic rocks forming part of the upper Paleozoic Samfrau geosyncline and is intruded by Mesozoic felsic plutons and dikes [*Craddock et al.,* 1964]. Three sites from Triassic to Jurassic plutons define a VGP at 82°S, 110°E, whereas eight sites from Jurassic to Cretaceous mafic dikes give a VGP at 86°S, 164°E [*McElhinny,* 1973, p. 307]. Both poles are also located near the present geographic South Pole.

Paleomagnetic data from Lower Cretaceous plutons from Marie Byrd Land [*Scharnberger and Scharon,* 1972] define a pole (36°S, 116°E) that is highly divergent from the VGP's of western Ellsworth Land and the Antarctic Peninsula. These data support the contention [*Elliot,* 1972; *Beck,* 1972; *Heirtzler,* 1971] that West Antarctica consists of at least two continental blocks with differing drift histories since Cretaceous time. *Scharnberger and Scharon* [1972] also suggest that part of West Antarctica (Marie Byrd Land), New Zealand, and the Campbell Plateau constituted a block that separated from Gondwanaland prior to Cretaceous sea floor spreading and subsequently broke apart during Late Cretaceous time by the initiation of spreading on the East Pacific Rise.

Test for Oroclinal Bending of the Antarctic Peninsula Since Cretaceous Time

A 30° change in orientation in the trends of fold axes along the Lassiter Coast parallels the apparent oroclinal bend of the peninsula in this area [*Kellogg and Rowley,* 1974]. Similar structural trends parallel to the axis of the peninsula are observed along its entire length [*Dalziel and Elliot,* 1972]. In past attempts to use paleomagnetic directions to determine whether oroclinal bending occurred after emplacement of the studied rocks, no limits of uncertainty in declination were included [*Hamilton,* 1966; *Dalziel et al.,* 1973].

In this report we have determined an uncertainty parameter for declination at each locality by calculating a confidence limit at the 95% level on the mean declination. This value (δ_{95}) is the angle subtended between the mean declination and the tangent to the cone of confidence (α_{95}) from the center of an equal-area projection and is equal to $\sin^{-1}(\sin \alpha_{95}/\cos I)$. Total uncertainty in the declination, then, is simply given by $2(\delta_{95})$. Confidence limits on declination are indeterminate if $I + \alpha_{95} > 90°$, inasmuch as the confidence cone includes the vertical.

Individual sample directions are seldom included in the paleomagnetic literature. For this reason the following method was used to calculate the uncertainties in declination for the two localities of the Lassiter Coast and for five other localities: Andean intrusives of the Trinity Peninsula, Argentine Islands, Stonington Island [*Blundell,* 1962], and Livingston Island [*Dalziel et al.,* 1973] and Cretaceous(?) dikes from western Ellsworth Land (C. K. Scharnberger, written communication, 1974). A weighted mean direction for a locality was calculated by considering each site mean direction N times (N being the number of samples at the site) and combining all directions from the locality to obtain the appropriate statistical parameters. Using the total number of samples from each locality in this way yields much lower values of α_{95} than would be derived by averaging the site means of a locality. (Both analyses give mean directions that are within a degree of each other.) The pertinent statistics from each locality are summarized in Table 3.

Within the confidence limits calculated for declination at each locality, however, no oroclinal bending can be unambiguously discerned (Figure 3). In fact, mean declinations and their confidence limits of three localities (Livingston Island, Stonington Island, and the Argentine Islands) lie to the west of north, opposite to the direction which is to be expected if structural bending had occurred since emplacement of the sampled rocks. These data may in part reflect the effects of local structure, nondipole paleomagnetic fields, and local magnetic anomalies.

The results from the southern Antarctic Peninsula and western Ellsworth Land are inconclusive. The large confidence limits in declination from the Lassiter Coast localities, despite the small values of α_{95}, are the result of nearly vertical mean directions. Similarly, the combination of steep inclinations and an α_{95} of 8° results in a large δ_{95} for western Ellsworth Land.

Conclusions

Paleomagnetic results from the Lassiter Coast and other localities in the Antarctic Peninsula indicate that there has been little latitudinal (north-south) movement of this province within the last 100 m.y. This is an important constraint for future plate tectonic reconstructions for the Antarctic Peninsula area.

It appears that the northern Antarctic Peninsula, as far south as about 68°S, is the approximate limit for which paleomagnetic data can be used to test hypotheses of oroclinal bending. The data from north of 68°S, when they are analyzed with respect to dispersion about declination, do not support the idea of substantial post-Late-Cretaceous structural bend-

TABLE 3. Summary of Paleomagnetic Data, Antarctic Peninsula

Locality	Reference	Location	Sites/Samples	$\bar{I}$	$\bar{D}$	α_{95}	δ_{95}
Northern Lassiter Coast	this study	73.1°S, 62.7°W	11/66	−80.7	354.7	3.6	22.9
Central Lassiter Coast	this study	74.3°S, 63.8°W	6/30	−84.3	0.8	2.7	28.3
Trinity Peninsula	*Blundell* [1962]	63.4°S, 57.0°W	4/17	−74.6	358.8	1.6	6.0
Argentine Islands	*Blundell* [1962]	65.0°S, 64.0°W	6/43	−78.6	350.9	1.7	8.6
Stonington Island	*Blundell* [1962]	67.4°S, 67.0°W	6/17	−77.3	347.6	2.3	10.5
Livingston Island	*Dalziel et al.* [1973]	62.7°S, 60.3°W	4/21	−71.2	336.6	2.8	8.7
Western Ellsworth Land	C. K. Scharnberger (written communication, 1974)	73°S, 96°W	6/?	−75	1	8	32.5

$\bar{I}$ is mean inclination, negative upward; $\bar{D}$ is mean declination, clockwise from true north; α_{95} is semiangle of the cone of 95% confidence level; and δ_{95} is uncertainty in declination, equal to $\sin^{-1}(\sin \alpha_{95}/\cos I)$. The statistical data, with the exception of the unpublished data of Scharnberger, were calculated by the authors.

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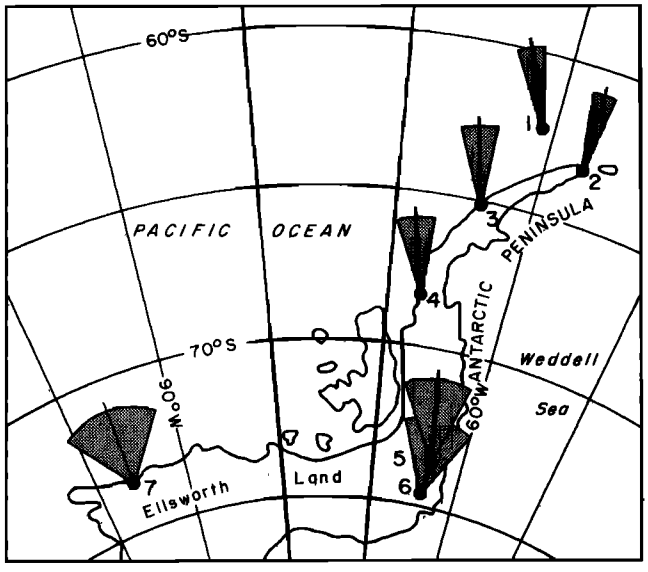


Fig. 3. A portion of West Antarctica showing the declination of mean paleomagnetic vectors and range of uncertainties (shaded) at the 95% confidence level. Sources of data: 1, *Dalziel et al.* [1973], intrusives from Livingston Island; 2, 3, and 4, *Blundell* [1962], Andean intrusives from Trinity Peninsula, Argentine Island, and Stonington Island, respectively; 5 and 6, this study, northern and central Lassiter Coast, respectively; and 7, C. K. Scharnberger (written communication, 1974), Cretaceous(?) intrusives.

ing. The possibility that some curvature of the southern part of the peninsula may have formed after Late Cretaceous time is not precluded by our data, nor do the paleomagnetic data deny the possibility that the peninsula has undergone post-mid-Mesozoic rigid rotation causing the gradual opening of the Weddell Sea, as has been proposed by *Heirtzler* [1971].

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